

Liquidity and Volatility Nexus in Cryptocurrency Markets

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Abstract

The growing inflow of funds in the crypto market makes understanding the dynamics and drivers of market liquidity a significant concern to drive informed regulatory, trading, and risk management decisions. This paper investigates the nexus between liquidity and volatility in cryptocurrency markets. We adopt the Asymmetric MEM-X, an extension of the well-known MEM, to model the dynamics of liquidity including realised volatility as exogenous variable at high frequency. This unified framework overcomes the shortcomings of traditional volatility and liquidity models which fail to assess the impact of market volatility on liquidity. In particular, the information about market liquidity is derived from easy-to-access transaction data through easy-to-compute bid-ask spread estimators. We apply the model to real data on two major cryptocurrencies, bitcoin versus US dollar (BTCUSD) and ethereum against US dollar (ETHUSD), considering two different exchanges, Bitstamp and Bitfinex. Our findings reveal (i) a distinct U-shaped intraday pattern in both liquidity and realized volatility, and (ii) a strong negative nexus between volatility and liquidity in cryptocurrency markets.

Keywords- Cryptocurrencies, Liquidity, Realised volatility, Multiplicative Error Model

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1 Introduction

Cryptocurrencies represent the most revolutionary breakthrough in the financial system, attracting considerable attention in the investment world as an alternative asset class. In recent years, the inclusion of cryptocurrencies in investors' portfolios to exploit attractive investment opportunities (Hong, 2017) and to improve diversification (Corbet et al., 2020, Borri, 2019) has flourished, generating trillions worth of trading and billions in open interest. After reaching a maximum of \$1.75 trillion in May 2021, today the total capitalisation of the crypto market is valued at \$1.05 trillion on a daily trading volume of \$38.02 billion¹. Cryptocurrencies are continuously traded 24 hours a day, including holidays and non-business days, in markets with a microstructure that highly departs from other asset classes. The transactions are directly cleared and settled across an immense multitude of centralised and decentralised exchanges in which the investors can efficiently transfer bitcoin for fiat and other (crypto)currencies². However, the rapid growth of the crypto ecosystem and its strong interconnection with traditional finance is not supported by a harmonized regulation designed to guarantee financial stability and market integrity. The blurred legislation under which exchanges operate does not protect investors adequately against risks associated with unsustainable business models, fraud, or market manipulations, leading to severe liquidity problems or price instability in the cryptocurrency market (Alexander and Heck, 2020).

In order to drive informed regulatory, trading, and risk management decisions in crypto markets, it is crucial to understand the dynamics of market liquidity, an elusive concept (Chordia et al., 2000) that can be deconstructed into three distinct dimensions: transaction costs, depth, and resiliency (Hasbrouck, 2007, Foucault et al., 2013). In this analysis, we focus on the transaction costs, measured by the bid-ask spreads, for at least two reasons. First, to model the liquidity formation in crypto markets, wherein prices are not fully integrated (Makarov and Schoar, 2020) and investors face constraints in redirecting orders and selecting the exchange for trading and hedging. Second, to describe the costs of illiquidity that may negatively impact the profits of investors who include cryptocurrencies in their investment strategies (Hu et al., 2019).

Nevertheless, a clear identification of factors influencing market liquidity has been relatively overlooked in the emerging literature primarily focusing on the risk-return relation (Zhang et al., 2021, Liu et al., 2022, Leong and Kwok, 2023). Since market makers tend to widen bid-ask spreads in response to increased market volatility, we consider volatility as a primary driver of liquidity. Market participants demand higher compensation to bear the heightened risks associated with a more volatile asset such as cryptocurrencies, known for their extreme price fluctuations (Corbet et al., 2018). Moreover, during crises, dealers prefer to trade less volatile assets that offer higher margins. This preference exacerbates liquidity spirals as more volatile assets experience a drying up of their liquidity (Brunnermeier and Pedersen, 2009, Malherbe, 2014).

We uncover the nexus between liquidity and volatility in cryptocurrency markets modeling this relationship within a unified econometric framework that attempts to explain the empirical regularities of two major cryptocurrencies³, bitcoin versus US dollar (BTCUSD) and ethereum against US dollar (ETHUSD), across two different exchanges. We consider Bitstamp, a CME Bitcoin Reference Rate⁴ constituent, and Bitfinex, a self-regulated exchange. In the context of liquidity and volatility modelling (see among others Engle and Gallo, 2006, Hafner et al., 2023,

¹Data are retrieved from <https://coinmarketcap.com/currencies/bitcoin/historical-data/>. The site was accessed on September 04, 2023.

²For further details about cryptocurrencies and blockchain mechanism we refer to Velde (2013), Yermack (2015) and Makarov and Schoar (2021)

³The market dominance of bitcoin and ethereum is currently 48.30% and 19.19%, respectively.

⁴CME Bitcoin Reference Rate is a daily reference rate of the U.S. dollar price of one bitcoin.

Alexander et al., 2022), we adopt a specification within the class of Multiplicative Error Model (MEM; Engle, 2002). These models are suitable to depict high-frequency dynamics for any non-negative financial variables, e.g. a liquidity estimator, described as the product of the conditional deterministic mean (scale factor) and the error term. In this framework, we model the dynamics of liquidity including realized volatility as an exogenous variable. We adopt a dynamic specification similar to that of Nguyen et al. (2020) to include the volatility variable in the conditional mean of the liquidity process, overcoming the baseline MEM's limited ability to account for the main features of liquidity. Hence, we adopt the Asymmetric MEM-X (AMEM-X) to model liquidity, combining the rich amount of volatility information from high-frequency exogenous variables on cryptocurrency with a standard MEM-type model that captures the sources of liquidity while preserving the main advantages of a multiplicative specification (Engle, 2002).

In the spirit of Brauneis et al. (2021a), we characterise the time series variation of liquidity on a cryptocurrency exchange using the Corwin and Schultz (2012) and Abdi and Rinaldo (2017) liquidity measures. These proxies, which are bid-ask spread estimators, offer similar performance compared to order book measures (such as Bid-Ask Spreads, Percentage Quoted Spread, Percentage Effective Spread, and Percentage cost of a roundtrip trade) while significantly reducing costs and processing time. In particular, the hourly liquidity measures are estimated from a dataset of transaction data (OHLC prices) sampled at a 1-minute frequency for bitcoin and ethereum on two large and reliable exchanges, Bitfinex and Bitstamp, over two years.

Our main finding is that the nexus between liquidity and volatility is strongly negative for BTCUSD and ETHUSD traded on the two exchanges. Specifically, a positive variation in volatility predicts a positive change in our liquidity variables, which corresponds to an increase in the bid-ask spreads and, in turn, to a deterioration of liquidity conditions in the next interval of time. Investors are attracted to the cryptocurrency market when volatility is less pronounced. This microstructure behaviour is explained by the decisions of market participants to protect their portfolios and to avoid the burden of higher transaction costs. By adjusting bid-ask spreads in response to volatility conditions, market makers induce a reduction in trading activity, thereby performing a stabilizing role. This result enriches the emerging literature on cryptocurrencies. Alexander et al. (2022) find that trading volume positively affects volatility and Brauneis et al. (2021a) show the positive relation between volume and bid-ask spreads. We provide instead a clear identification of the bridge between liquidity and volatility within a unified econometric model. In contrast to conventional models focused solely on either volatility dynamics or liquidity, such as the MEM (Engle, 2002) or DArLiQ (Hafner et al., 2023), we adopt an extension of a well-known MEM framework for high-frequency to model the nexus between volatility and liquidity. Furthermore, the inclusion of a new set of information enables the AMEM-X to capture better the variability of liquidity time series relative to the baseline MEM models. Our work also aligns with several empirical analyses for both traditional and cryptocurrency markets such as Chordia et al. (2005), Brauneis et al. (2021b) and Corbet et al. (2022).

A further contribution of our paper highlights a remarkable intraday pattern of liquidity and realised volatility for bitcoin and ethereum. During the trading day, the two variables follow a clear U-shape with several distinct 4-hourly spikes showing an intra-day pattern analogous to that of the Deutsche Mark/USD exchange rate documented in Andersen and Bollerslev (1997). Alexander et al. (2022) relate these spikes to the funding payments on perpetuals, after which some investor adjust their portfolios or exploit arbitrage opportunities between spot and perpetuals.

Our findings help a broad range of market participants to understand the dynamics and the factors influencing the market liquidity. Traders and fund managers, who seek to strategically time market liquidity for optimal entry or exit, can include information on the impact of transaction cost on their returns in the investments strategies. Regulators aiming to guarantee market

stability and strengthen the legislative infrastructure on cryptocurrencies can leverage our empirical results for better guidance of market liquidity evolution under periods of high volatility. Lastly, scholars can use our analysis as a background framework for further studies on crypto liquidity embedding volatility as the main driver.

The remainder of this paper is organised as follows: Section 2 outlines the econometric model for the liquidity process with the inclusion of the volatility variable; Section 3 presents the data and focuses on descriptive analysis of realised volatility and liquidity dynamics, highlighting their intraday pattern; Section 4 presents the empirical results showing the negative nexus between volatility and liquidity. Section 5 concludes.

2 Liquidity Variables

To characterise the time-series variation of liquidity in cryptocurrency markets, we rely on two liquidity proxies, the [Corwin and Schultz \(2012\)](#) and [Abdi and Ranaldo \(2017\)](#) measures, which are used as estimators of bid-ask spread. Such measures capture one of the dimensions of liquidity using a set of easy-to-acquire data⁵, the transaction cost of immediately buying or selling a security. Specifically, [Brauneis et al. \(2021a\)](#) evidences the superior performance of these transaction-based proxies in capturing the evolution of liquidity in cryptocurrency markets, emphasising enormous advantages in terms of cost and process time savings with respect to high-frequency order book measures (e.g., Bid-Ask Spreads, Percentage Quoted Spread, Percentage Effective Spread, Percentage cost of a roundtrip trade).

The [Corwin and Schultz \(2012\)](#) estimator is computed as

$$CS_{j,j+1} = \frac{2 (\exp(\alpha) - 1)}{1 + \exp(\alpha)}, \quad (1)$$

$$\alpha = \frac{\sqrt{2\beta} - \sqrt{\beta}}{3 - 2\sqrt{2}} - \sqrt{\frac{\lambda}{3 - 2\sqrt{2}}}, \quad \beta = \left[\ln \left(\frac{H_j}{L_j} \right) \right]^2 + \left[\ln \left(\frac{H_{j+1}}{L_{j+1}} \right) \right]^2, \quad \lambda = \left[\ln \left(\frac{H_{j,j+1}}{L_{j,j+1}} \right) \right]^2,$$

where H_j and L_j are the high and low prices in the 1-minute interval j within a trading hour t , $H_{j,j+1}$ and $L_{j,j+1}$ denote respectively the high and low prices over two adjacent subintervals j and $j + 1$ within t . We set all the negative values to zero. Then, the CS_t estimator for interval t is computed as the average of the CS measures for all subintervals in each t .

The [Abdi and Ranaldo \(2017\)](#) 'two-day corrected' estimator is

$$AR_j = \sqrt{\max [4 (c_j - \bar{p}_j) (c_j - \bar{p}_{j+1})]}, \quad (2)$$

$$h_j = \ln(H_j), \quad l_j = \ln(L_j), \quad c_j = \ln(C_j), \quad \bar{p}_j = \frac{(h_j + l_j)}{2},$$

where h_j, l_j, c_j are the natural logarithms of high, low and close prices, respectively, in subinterval j , while $\bar{p}_j$ is the midpoint between the h_j and l_j . For period t , the AR_t estimator is the mean of the AR estimators across all adjacent subintervals in t , $AR_t = \frac{1}{J-1} \sum_{j=1}^J AR_{t,j}$.

⁵Calculating bid-ask spreads typically involves the use of high-frequency data, which can be both costly to obtain and power-demanding to process.

3 The Model

We consider a non-negative valued process x_t , the liquidity at the time t . Following [Engle \(2002\)](#), we assume that the evolution of x_t can be specified by the Asymmetric Multiplicative Error Model (AMEM) as

$$x_t = \mu_t \varepsilon_t \quad \varepsilon_t | \mathcal{F}_{t-1} \stackrel{iid}{\sim} D^+(1, \sigma_\varepsilon^2) \quad (3)$$

with

$$\mu_t = \omega + \sum_{i=1}^q \alpha_i x_{t-i} + \sum_{i=1}^p \beta_i \mu_{t-i} + \gamma x_{t-1}^-, \quad (4)$$

where $\mathcal{F}_{t-1}$ is the σ -field generated by $\{x_0, \dots, x_{t-1}, \mu_0\}$, μ_t is the deterministic conditional expectation of the process x_t measurable w.r.t $\mathcal{F}_{t-1}$, $x_{t-1}^- = x_{t-1}$ if the return of the respective interval is negative and zero otherwise, γ is the parameter that captures the asymmetric response of the x_t to upcoming news (i.e leverage effect), and p and q determine the lag structure. The AMEM preserves the positivity of the variable of interest without resorting to logs. However, it requires restrictions on parameters to guarantee the positivity of μ_t and the covariance stationarity of the process, x_t , with probability one. We require $\omega > 0, \alpha_i \geq 0, \beta_i \geq 0 \quad \forall i$ and $\sum_{i=1}^p \alpha_i + \sum_{i=1}^q \beta_i + \frac{\gamma}{2} < 1$.

We estimate the model via the Quasi-Maximum Likelihood (QML) approach assuming that the innovations ε_t are given by a sequence of i.i.d random variables which follow a distribution with strictly positive support and unit mean. We use Gamma distribution, $\Gamma(\frac{1}{\theta}, \theta)$, with density function

$$f_\varepsilon(\varepsilon_t | \mathcal{F}_{t-1}) = \frac{1}{\varepsilon_t} (\varepsilon_t \theta)^\theta \frac{1}{\Gamma(\theta)} e^{-\theta \varepsilon_t}, \quad (5)$$

where $\frac{1}{\theta}$ is the scale and θ is the shape parameter. Therefore, $\mathbb{E}[\varepsilon_t | \mathcal{F}_{t-1}] = 1$ and $\mathbb{V}[\varepsilon_t | \mathcal{F}_{t-1}] = \frac{1}{\theta}$. Consequently, $\mathbb{E}[x_t | \mathcal{F}_{t-1}] = \mu_t$ and $\mathbb{V}[x_t | \mathcal{F}_{t-1}] = \frac{\mu_t^2}{\theta}$ allow the AMEM to accommodate the heteroskedastic effects frequently observed in financial time series.

Despite the robustness of MEMs to the presence of measurement errors ([Cipollini et al., 2021](#)), the baseline AMEM shows a limited ability to characterize the evolution of realized measures. Indeed, the correct specification of μ_t plays a crucial role in describing the rich dynamic behaviour of financial time series. Therefore, we consider the AMEM-X which includes information provided by a weakly exogenous variable z_{t-1} , the realised volatility, in the conditional mean of liquidity variable x_t

$$x_t = \mu_t \varepsilon_t \quad \varepsilon_t | \mathcal{F}_{t-1} \stackrel{iid}{\sim} \Gamma(1, \frac{1}{\theta}) \quad (6)$$

with

$$\mu_t = \omega + \sum_{i=1}^p \alpha_i x_{t-i} + \sum_{i=1}^q \beta_i \mu_{t-i} + \gamma x_{t-1}^- + \delta z_{t-1}, \quad (7)$$

where we do not require any condition on δ to ensure the covariance stationarity of x_t and the positivity of μ_t .

4 Data: Prices, Realised Volatility and Liquidity Measures

The crypto asset and derivatives market evolves rapidly and acquires growing importance for payments and investment portfolios while presenting a microstructure that differs significantly from other asset classes. Investors can trade cryptocurrencies on numerous self-regulated, centralized and decentralized exchanges with unique characteristics. They are open 24 hours a day, including holidays and non-business days. They allow the transfer, clearing and settlement of bitcoin against various fiat and other cryptocurrencies directly using bank accounts or credit cards.

While trading and innovations have flourished, these markets are still highly fragmented and weakly regulated, making it difficult for investors to choose a specific exchange or find restrictions to redirect orders to more convenient platforms. However, the regulators in the USA, China and mainly the EU are actively taking measures to preserve market integrity and financial stability while providing consumers with essential information about cryptocurrency risk⁶.

We collect minute-by-minute historical transaction data⁷ of two exchanges, Bitstamp and Bitfinex, for two major cryptocurrencies, bitcoin versus US dollar (BTCUSD) and ethereum against US dollar (ETHUSD), from 2020-11-16 00:00:00 UTC to 2022-03-16 00:00:00 UTC, a total of 485 trading days with 1690814 observations. The data set consists of 1-minute open, high, low and close (OHLC) prices, UNIX timestamps and volume in USD. We focus on the recent bull market of the crypto market, where the price of bitcoin and ethereum has rapidly evolved presenting significant oscillations. Bitcoin price reached a historical maximum in 2021, surpassing 65,000 USD in November (Figure 1) driven by the introduction of the first Bitcoin ETF, Tesla’s purchase of 1.5 billion U.S. dollars in Bitcoin, and IPO of the largest cryptocurrency exchange in the United States. Similar to Bitcoin, the price of ETH experienced a similar pattern reaching around 4,800 USD in 2021 (Figure 2). This increase was attributed to technological developments, including the Berlin update, along with the sale of NFTs for 38,000 ETH.

The selection of the exchange is driven by the analysis of [Hougan et al. \(2019\)](#) and [Le Pennec et al. \(2021\)](#) which highlight the importance of the reliability of the reported transactions. They analyse granular data, which is not available via the exchanges’ APIs, and find that the operators tend to report not economically meaningful or fake transactions to inflate trading volume in crypto markets. After exploiting several identification tests, Bitstamp and Bitfinex are indicated as “real volume” and reliable exchanges.

It is possible to detect missing data due to technical problems, exchange-specific trading halts and missing trading activity. For this reason, when we aggregate data from 1-minute to hourly frequency, we require the availability of 80% of valid data for each interval. In particular, this corresponds to 48 minutes for each trading hour⁸, t . Our resulting data set contains approximately 11640 trading hours.

⁶In particular, the European Union adopts on 20 April 2023 the Markets in Crypto-Assets Regulation (MiCA), establishing a harmonized legislative framework for crypto-assets that fall outside the scope of existing financial services legislation. MiCA is the world’s first and only such regulation. Pursuing a comprehensive legislative framework in the USA faces significant obstacles due to the battle between the SEC and CFTC for control over the cryptocurrency landscape. Several drafts have been proposed, including the Digital Asset Market Structure and Investor Protection Act.

⁷The transaction data can be purchased from data providers, such as CoinAPI or Kaiko, or freely downloaded from cryptodatadownload.com.

⁸According to coinmarket.com, a trading hour is defined as a time span of 59 minutes.

Figure 1: Bitcoin Price



Note: The figure shows the bitcoin price on Bitstamp from 6 November 2020 to 16 March 2022.

Figure 2: Ethereum Price



Note: The figure shows the ethereum price on Bitstamp from 6 November 2020 to 16 March 2022.

When it comes to the analyses of the realized volatility in a high-frequency framework, the estimators should account for the market microstructure noise. This is the result of “micro” frictions in the trading process (e.g. price discreteness, bid-ask bounces, etc.) that subsequently translates into “macro” consequences relative to asset prices’ liquidity ([Ait-Sahalia and Yu, 2008](#); [Seifoddini et al., 2017](#)). [Andersen \(2000\)](#) shows that increasing the frequency of data allows for better capturing of volatility flows at the cost of higher microstructure noise. Reasonably, we select a 1-minute data sampling frequency from which we compute hourly realised volatility. The selection of minute-by-minute aggregation is a more convenient alternative relative to a higher frequency (e.g. tick-by-tick, second-by-second) in terms of financial resources and data management. Indeed, to compute a noise-robust estimator of realised volatility using high-frequency data, we use the pre-averaging approach presented by [Jacod et al. \(2009\)](#). Here, the impact of microstructure noise is locally smoothed by applying a weighted average to a one-minute log return. The pre-averaged log returns are given by

$$\bar{r}_j = \sum_{l=1}^{k_n} g\left(\frac{l}{k_n}\right) r_{j+l} \quad (8)$$

$$r_j = \log\left(\frac{C_j - O_j}{O_j}\right),$$

where C_j is the closing price, O_j is the opening price, r_j is the ordinary (not pre-averaged) log returns and g is a tent-shaped weighting function, $g(x) = \min(x, 1 - x)$ ⁹. The k_n bandwidth of the local pre-averaging window is inversely related to the sampling frequency (Hautsch and Podolskij, 2013). Specifically, it is optimally set at $[\theta\sqrt{n}] = 3$, where n is the number of subintervals j in i (in our case $n = 60$); $\theta \in [0.3, 0.6]$ denotes the smoothing parameter and depends on the frequency of the data and the liquidity of the asset. Following Hautsch and Podolskij (2013), we choose θ as 0.4 a value that allows us to avoid the oversmoothing effect and negative bias in the estimator. The hourly pre-averaging realized variance estimator is obtained by summing up squared pre-averaged 1-minute returns over each trading hour. Finally, we compute the pre-averaged realised volatility as the square root of the realised variance.

Table 1 and 2 report descriptive statistics on liquidity measures, Corwin and Schultz (2012) and Abdi and Rinaldo (2017) estimators¹⁰, and realized volatility for the pair BTCUSD and ETHUSD over the period from 16 November 2020 to 16 March 2022. The liquidity measures in Abdi and Rinaldo (2017) and Corwin and Schultz (2012) suggest that ETHUSD is less liquid than BTCUSD indicating a lower investor willingness to trade and a consequent higher transaction cost. This evidence is supported by the volatility measures. The average levels of volatility are significantly higher for ETHUSD than for BTCUSD on all exchanges. We also notice high values for the standard deviation of ETHUSD highlighting strong oscillations around the mean. These results are due to the different levels of trading activity which are subsequently connected to macroeconomic phenomena or investor sentiment.

Table 1: Liquidity measures and Realized Volatility for BTCUSD

Exchanges		Mean	Std	25%	50%	75%	no. hourly obs
Bitstamp	CS	2.4246	2.2144	1.0959	1.8900	3.0562	11640
	AR	4.4101	2.5390	2.8793	3.8568	5.2314	
	RVol	0.0037	0.0026	0.0022	0.0030	0.0043	
Bitfinex	CS	2.0937	2.0778	0.8802	1.5288	2.5758	11524
	AR	3.5434	2.4615	1.9721	2.9358	4.3804	
	RVol	0.0037	0.0026	0.0021	0.0030	0.0043	

Note: The table shows the descriptive data on liquidity measures, Abdi and Rinaldo (2017) and Corwin and Schultz (2012) estimators expressed in basis points, and realized volatility (RVol) for the pair BTCUSD at hourly frequency over the period from 16 November 2020 to 16 March 2022.

Following the empirical analysis of Engle et al. (2013), Li et al. (2020) and Alexander et al. (2022), to diminish the influence of outliers, we winsorize the top 0.05% of realised volatility and liquidity values, computed according to (1)-(8), setting all the observations greater than the 99.95%-quantile equal to this quantile.

⁹The real-valued function $g : [0, 1] \rightarrow \mathbb{R}$ must be continuous, piecewise continuously differentiable with piecewise Lipschitz derivative and having $g(0)=g(1) = 0$.

¹⁰We remind you that Abdi and Rinaldo (2017) and Corwin and Schultz (2012) are both estimators of the bid-ask spread, which is an indicator of the liquidity in terms of the transaction cost. In particular, more liquid stocks tend to have tighter spreads.

In contrast to most theoretical predictions, a positive relation between volume and bid-ask spreads is documented both for large US stocks (Bogousslavsky and Collin-Dufresne, 2019) and cryptocurrencies (Brauneis et al., 2021a). Moreover, Alexander et al. (2022) revealed a strong positive volume-volatility relationship for crypto markets.

Table 2: *Liquidity measures and Realized Volatility for ETHUSD*

Exchanges		Mean	Std	25%	50%	75%	no. hourly obs
Bitstamp	CS	1.6833	2.5190	0.4058	0.9513	2.0451	11640
	AR	4.4973	3.4983	2.3027	3.6965	5.6718	
	RVol	0.0041	0.0033	0.0022	0.0032	0.0049	
Bitfinex	CS	2.5053	2.6475	1.0148	1.7535	3.0465	11507
	AR	4.6235	3.2670	2.6415	3.8236	5.5839	
	RVol	0.0045	0.0031	0.0027	0.0037	0.0053	

Note: The table shows the descriptive data on liquidity measures, Abdi and Rinaldo (2017) and Corwin and Schultz (2012) estimators expressed in basis points, and realized volatility (RVol) for the pair ETHUSD at hourly frequency over the period from 16 November 2020 to 16 March 2022.

Consistently, we notice for the two major cryptocurrencies a strong correlation between liquidity and volatility dynamics. Specifically, we find that high volatility leads to high illiquidity. This phenomenon is explained by (i) the liquidation of crypto portfolios following extreme events for a future market positioning and (ii) the action of the market makers that increases the bid-ask spreads and contributes to the tightening of liquidity conditions. Figure 3 shows this empirical evidence for BTCUSD and ETHUSD pairs reporting the dynamics of volatility for the selected sample period together with the Corwin and Schultz (2012) and Abdi and Rinaldo (2017) estimators. Finally, we observe the clustering phenomena for the dynamics of liquidity and volatility.

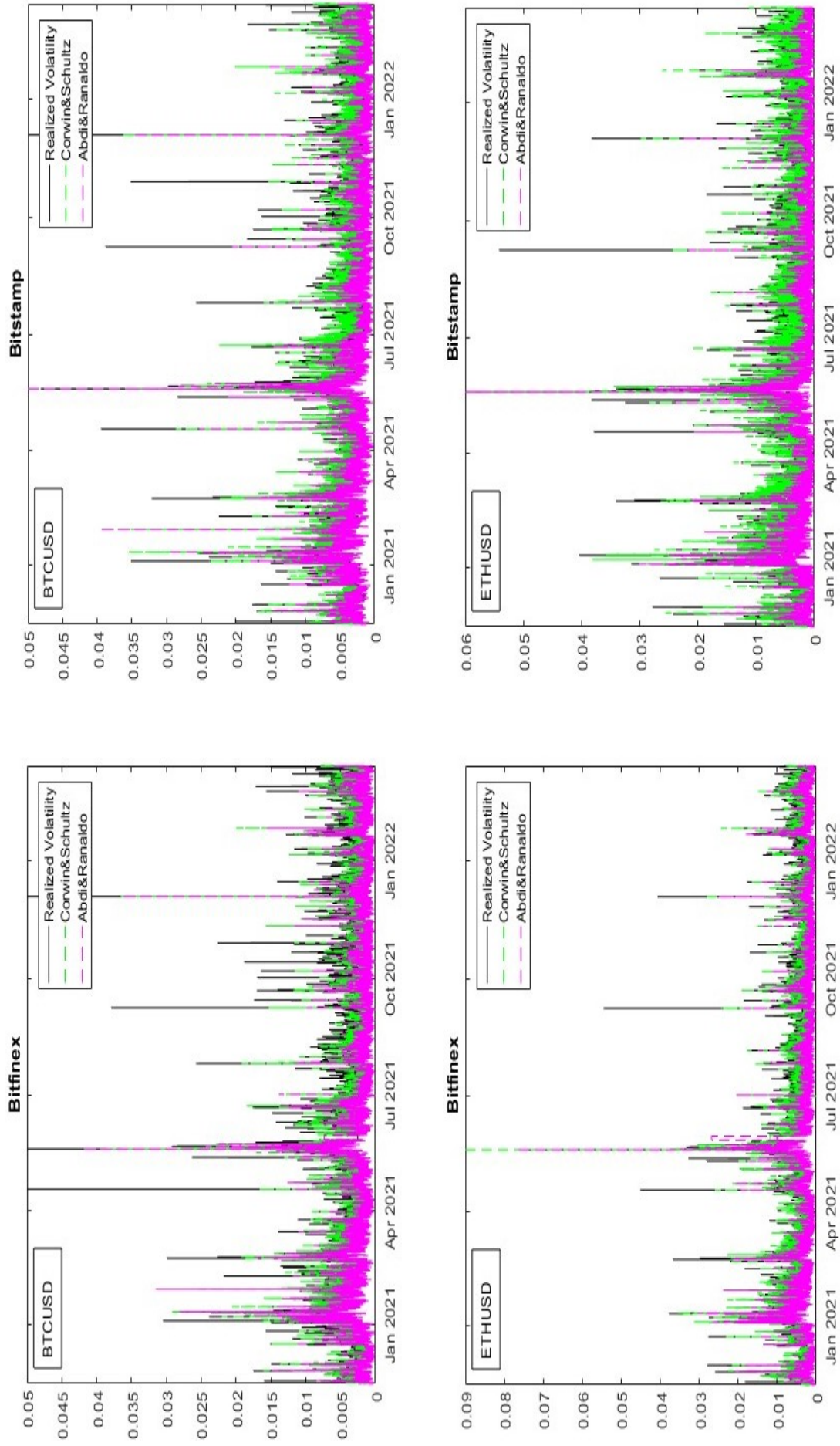
The graphs of Figure 5 show another interesting feature. We document an intra-day pattern of liquidity and realized volatility for both BTCUSD and ETHUSD analogous to that of the Deutsche Mark/USD exchange rate documented by Andersen and Bollerslev (1997). They show that the average variation in market liquidity and volatility follows a U-shape pattern, characterized by clear peaks occurring at midnight, 14:00, and 16:00 UTC, along with smaller spikes at 08:00 UTC. After 16:00, the volatility and liquidity trail off interrupted by another distinct spike at 20:00 UTC.

In our empirical analysis, we account for the intra-day pattern following Alexander et al. (2022) that adopts a non-parametric approach by dividing each observed raw variable by the average of the corresponding hourly interval over the entire sample period, avoiding an exponential increase in the number of parameters in the MEM.

5 Empirical Results

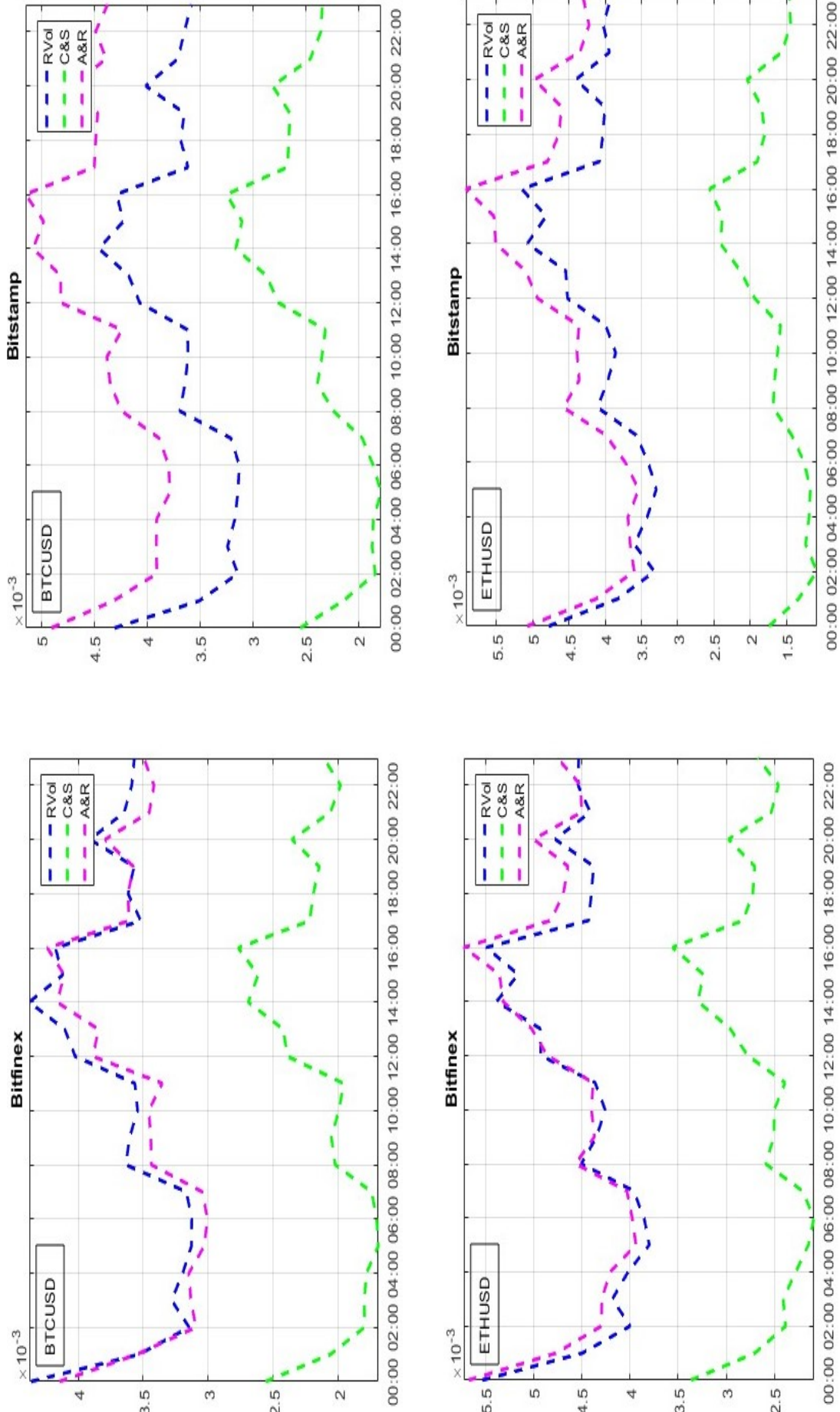
To model the effect of a change in the crypto exchange’s volatility on the liquidity dynamics, we estimate the AMEM-X in which we set $p = q = 1$ for simplicity. In particular, this choice is driven by the desire to follow parsimonious criteria for which the preferred model is the one that produces the best fit with the least number of parameters, thus reducing the randomness in the final outcome. Then, we compare the estimates of our model with the baseline AMEM (Engle, 2002) specified as in (3) - (4). Table 3 and 4 show the parameter estimates on Bitstamp and Bitfinex for BTCUSD and ETHUSD together with the Bayesian Information Criterion (BIC), log-likelihood (LLK) and t-stat for Likelihood ratio test (LR). Overall, the liquidity patterns show significant similarities across exchanges and instruments. The estimates of α and β are all statistically significant at 1% level and present similar values but higher for those estimated by AMEM due to the absence of the realised volatility in the conditional mean specification of the liquidity

Figure 3: Bitcoin and Ethereum Liquidity Estimators and Realized Volatility



Note: The figure shows the Abdi and Rinaldo (2017) and Corwin and Schultz (2012) liquidity measures, rescaled by a factor of 10, and realized volatility on Bitfinex for the pair ETHUSD at hourly frequency over the period from 6 November 2020 to 16 March 2022.

Figure 4: Bitcoin and Ethereum Volatility and Liquidity Intra-day Pattern



Note: Intraday patterns of Corwin and Schultz (2012), Abdi and Rinaldo (2017) and realised volatility variables. The liquidity proxies are rescaled by a factor of 10. For each hour interval of the trading day, we divide each variable of interest by the average of the respective hour interval computed over the complete period from 6 November 2020 to 16 March 2022.

process. The total persistence - the sum of α and β - of hourly liquidity varies from 0.9005 to 0.9349 for the AMEM. It ranges from 0.6505 to 0.8380 for all the specifications in the AMEM-X, indicating less persistence of μ_t due to the effect of the realised volatility. The low persistence compared to conventional autoregressive models is the result of the partial cleaning of the series from autocorrelation. More specifically, we show that the coefficients α and β , which capture short- and long-term persistence, significantly decrease in all models once we account for the realised volatility. In other words, the role of past liquidity loses importance in predicting future liquidity when volatility is considered. A positive variation in volatility predicts a positive change in the liquidity variables, corresponding to an increase in the bid-ask spreads, which determines a deterioration of liquidity conditions in the next hourly interval. To understand how much liquidity is explained by the volatility, we analyse the δ parameter that ranges from 0.1324 to 0.2768. This parameter indicates that an increase in the volatility measure results in a rise of 12%-30% in liquidity for the subsequent hourly interval. Moreover, α and β suggest that the ETHUSD pair traders are generally more reactive to current market conditions than BTCUSD. One possible explanation could be attributed to trading activity driven by informed investors who are very sensitive to the disclosure of new information or to changing market conditions. Unsurprisingly, the γ parameter estimates are positive and significant on all exchanges. These results document the leverage effect capturing the asymmetric response of the liquidity to upcoming news often observed in other asset classes. The magnitude of this effect remains notably consistent across various instruments, ranging from 0.0148 to 0.0588. This indicates that a negative return leads to an increase of approximately 1 to 6% in the liquidity for the next hourly interval.

According to the Bayesian information criterion (BIC) and the likelihood ratio test, the AMEM-X provides superior performance relative to AMEM in explaining the dynamics of [Corwin and Schultz \(2012\)](#) or the [Abdi and Rinaldo \(2017\)](#) after the inclusion of the realised volatility, across the different exchanges and cryptocurrency. We notice that the estimates of AMEM-X describe the variability in the data and better capture the magnitude of the large variations in the evolution of liquidity. Figure 5 and 6 present the observed liquidity, measured by [Corwin and Schultz \(2012\)](#) or the [Abdi and Rinaldo \(2017\)](#), for BTCUSD and ETHUSD pairs, respectively, and the estimated liquidity using the AMEM-X with the inclusion realised volatility.

Table 3: *Parameter estimates for BTCUSD*

Exchange	Parameters	AMEM ^{CS}	AMEM ^{CS} -X	AMEM ^{AR}	AMEM-X ^{AR}
Bitstamp	ω	0.0454***	0.1447***	0.0484***	0.3276***
	α	0.4297***	0.3941***	0.3320***	0.2059***
	β	0.5051***	0.4439***	0.6029***	0.4545***
	γ	0.0441***	0.0379***	0.0336***	0.0281***
	δ	-	0.1324***	-	0.2437***
	θ	4.6563***	4.7393***	7.9999***	8.3630***
	BIC	9160.75	8940.70	6197.63	5659.99
	LLK	-4580.38	-4580.38	-3098.82	-2829.99
	LR		220.06***		537.65***
Bitfinex	ω	0.0696***	0.2230***	0.0617***	0.3141***
	α	0.4334***	0.3711***	0.3926***	0.2525***
	β	0.4671***	0.3706***	0.5180***	0.4088***
	γ	0.0532***	0.0466***	0.0523***	0.0461***
	δ	-	0.1826***	-	0.2681***
	θ	3.4415***	3.5006***	5.7948***	6.0038***
	BIC	11507.39	11290.47	11507.39	8108.31
	LLK	-5753.69	-5645.23	-5753.69	-4054.15
	LR		216.92***		3399.08***

Note: The table reports the coefficients of the different MEM model specifications estimated via QMLE for two exchanges, Bitfinex and Bitstamp. *, **, *** represents the significance at 10%, 5% and 1%, respectively. BIC is the Bayesian information and criterion; LLK is log-likelihood and LR is the t-stat for Likelihood ratio test.

Table 4: *Parameter estimates for ETHUSD*

Exchange	Parameters	AMEM ^{CS}	AMEM-X ^{CS}	AMEM ^{AR}	AMEM-X ^{AR}
Bitstamp	ω	0.0688***	0.2151***	0.0610***	0.2983***
	α	0.4975***	0.4351***	0.4091***	0.2730***
	β	0.4160***	0.3265***	0.5133***	0.4223***
	γ	0.0412***	0.0264**	0.0275***	0.0148**
	δ	-	0.1324***	-	0.2612***
	θ	1.3742***	1.3944***	4.5665***	4.7014***
	BIC	14492.11	14284.98	10358.40	9995.47
	LLK	-7246.05	-4580.38	-5179.20	-4997.73
	LR		207.14***		362.93***
Bitfinex	ω	0.0642***	0.2465***	0.0519***	0.3228***
	α	0.4586***	0.4089***	0.3829***	0.2395***
	β	0.4449***	0.2990***	0.5393***	0.4110***
	γ	0.0588***	0.0476***	0.0518***	0.0525***
	δ	-	0.2109***	-	0.2768***
	θ	3.2555***	3.3245***	5.3908***	5.5520***
	BIC	10717.44	10448.79	9190.23	8826.45
	LLK	-5358.72	-5224.40	-4595.12	-4413.23
	LR		268.6507***		363.78***

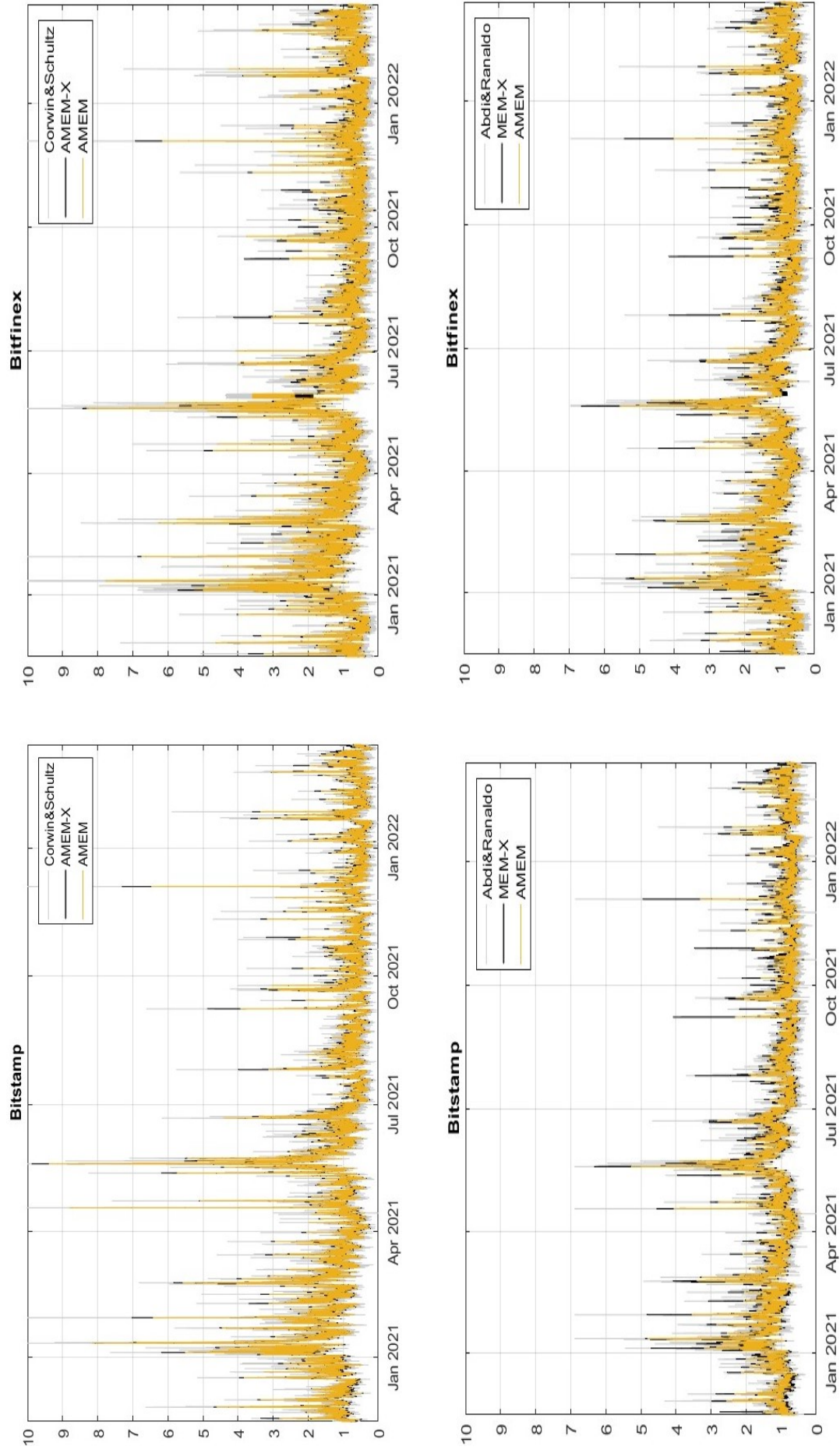
Note: The table reports the coefficients of the different MEM model specifications estimated via QMLE for two exchanges, Bitfinex and Bitstamp. *, **, *** represents the significance at 10%, 5% and 1%, respectively. BIC is the Bayesian information and criterion; LLK is log-likelihood and LR is the t-stat for Likelihood ratio test.

6 Conclusions

We investigate the liquidity and volatility nexus in cryptocurrency markets. To do so, we adopt the Asymmetric MEM-X, an extension of the well-known MEM (Engle, 2002), in which we model the dynamics of liquidity including realised volatility as exogenous variables, similar to Nguyen et al. (2020). Our model combines information from a high-frequency volatility variable with the MEM-type model to capture sources of liquidity while preserving the main advantages of a multiplicative specification.

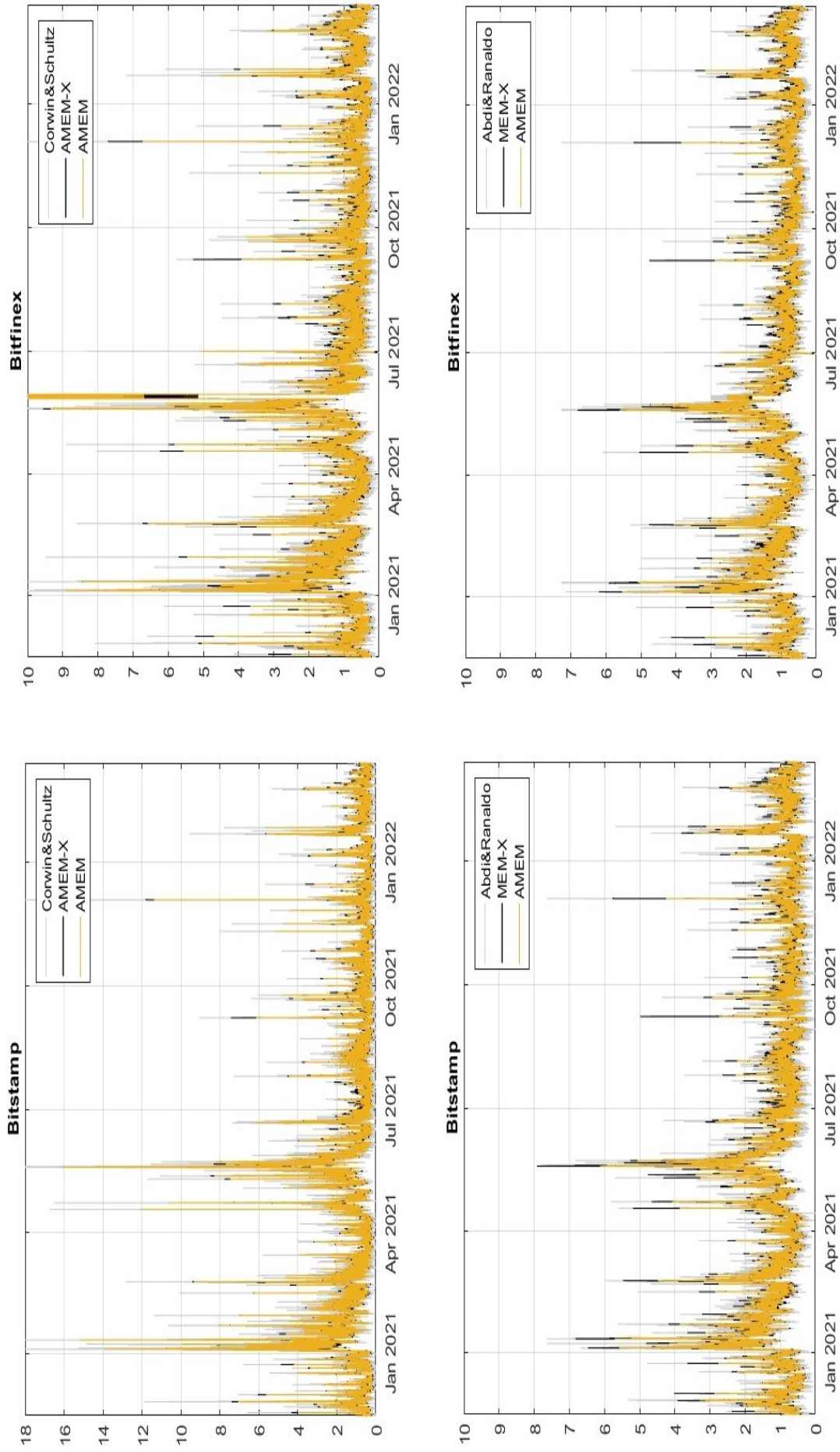
Our analysis covers minute-by-minute transaction data of two major cryptocurrencies, bitcoin versus US dollar (BTCUSD) and ethereum against US dollar (ETHUSD), across two different exchanges, Bitstamp and Bitfinex, over two years. We characterize the time-series variation of liquidity using two bid-ask spread estimators, Corwin and Schultz (2012) and Abdi and Rinaldo (2017), which provide similar performance while significantly reducing costs and processing time compared to order book measures (Brauneis et al., 2021a). Subsequently, we estimate the AMEM-X, and we observe that the nexus between liquidity and volatility is strongly negative for BTCUSD and ETHUSD traded on the two exchanges. Specifically, a positive variation in volatility predicts a positive change in our liquidity variables, which corresponds to an increase in the bid-ask spreads. The resulting deterioration of liquidity conditions leads to a reduction in trading activity, highlighting a stabilizing function of market makers who adjust bid-ask spreads in response to market conditions. These results enhance the recent evidence in Alexander et al. (2022) showing that trading volume positively affects volatility, in Brauneis et al. (2021a) showing a positive relation between volume and bid-ask spreads, and in several empirical analyses such as Chordia et al. (2005), Brauneis et al. (2021b) and Corbet et al. (2022). We provide a clear identification of the nexus between liquidity and volatility with a unified econometric framework. Unlike traditional models that only explain the dynamics of volatility and liquidity separately, such as MEM (Engle, 2002) or DARLiQ (Hafner et al., 2023), we focus on the joint modelling of two variables of interest providing a superior tool for explaining their interaction. Furthermore, the inclusion of a new set of information enables the AMEM-X to capture better the variability

Figure 5: Liquidity Estimation for BTCUSD



Note: The figure shows the bitcoin liquidity (grey line), measured by Corwin and Schultz (2012) or Abdi and Rinaldo (2017) estimator, and the AMEM-X (black line) and AMEM (yellow line) estimated liquidity, including realised volatility, on Bitstamp and Bitfinex.

Figure 6: Liquidity Estimation for ETHUSD



Note: The figure shows the bitcoin liquidity (grey line), measured by Corwin and Schultz (2012) or Abdi and Rinaldo (2017) estimator, and the AMEM-X (black line) and AMEM (yellow line) estimated liquidity, including realised volatility, on Bitstamp and Bitfinex.

of liquidity time series relative to the baseline MEM models.

Our analysis also offers robust empirical evidence supporting the existence of a distinct U-shaped intraday pattern in both liquidity and realized volatility for Bitcoin and Ethereum, which has not been extensively documented in the emerging literature.

However, the empirical application shows that AMEM-X partially accounts for the extreme movements observed in the liquidity dynamics of cryptocurrencies. This is left to future research in which our goal is to enhance the existing model, as suggested by [Caporin et al. \(2017\)](#), by incorporating a multiplicative jumps term to capture and evaluate unpredictable large movements effectively.

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